Asher Olgeirson

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Education

B.S. Computer Engineering and Computer Science *Aug 2025 - May 2029 (expected)*

Northeastern University, Boston, MA GPA 3.4

Relevant Coursework: Fundamentals of Networks, Embedded Design Enabling Robotics, Computing Fundamentals for Engineers, Discrete Structures, Program Design and Implementation, Physics I & II

Technical Skills and Interests

Languages: C, C++, Python, Rust, Go, Java
Embedded: CAN, I2C, embassy-rs, ThreadX (RTOS), STM32, ESP32
Systems & Tools: Linux, Git, Docker, PostgreSQL, SQLite, Grafana, GDB, AWS, Pytest
Areas of Interest: Embedded Systems, Autonomy, Robotics

Experience

Northeastern Electric Racing *Jan 2026 - Present*

Firmware Engineer, Boston, MA

– Rewrote safety-critical car status subsystem in Rust using thread-safe embassy-rs
– Implemented a time-of-flight sensor driver that communicates car configuration data over I2C and CAN
– Debugged and helped develop thermal management systems for critical components
– Streamlined DevOps tasks using Python and Docker

University College Dublin Formula Student *Sep 2025 - Dec 2025*

Powertrain Engineer, Dublin, Ireland

– Wrote safety-critical, fault-tolerant throttle-by-wire system in C++
– Developed C++ interface to control the primary motor over CAN

Comcast Corporation *Summer 2023-2025*

Software Engineering Intern, Denver, CO

– Returned for three consecutive summers, each on a different engineering team
– Built developer tooling to analyze JWT, ASN.1, and custom XML tokens in Rust (2025)
– Automated sprint visualization by generating Mermaid diagrams from Jira APIs in Rust (2025)
– Developed a device-identification system in Go used by Comcast, Liberty Mutual, Boeing, and Amtrak (2024)
– Engineered secondary monitoring for fiber-optic infrastructure using Java, PostgreSQL, and Grafana (2023)

Brute Force Robotics (FRC 2240) *Aug 2022 - Apr 2025*

Software Captain, Denver, CO

– Led 10-person software team to 2 appearances at the FIRST World Championship
– Designed, wrote, and tested embedded robot software in C++ over 8-week build seasons
– Created platforms for full and partial autonomy using gyroscopic navigation, computer vision, and machine learning
– Migrated the codebase to a declarative, lambda-based state-machine framework
– Fused camera, gyroscope, and wheel-sensor data with a Kalman filter for odometry

Engineering Projects

SushiRoll *Sep 2026 - Present*

– Built an Android app in Kotlin that turns the phone into a mouse using its accelerometer
– Computed pointer deltas by integrating gravity-compensated acceleration with zero-velocity updates (ZUPT)

Margin *Aug 2026*

– Built a math-problem autograder with LaTeX input, rendering, and output
– Generated feedback using platform-agnostic LLMs via OpenRouter

Garfana (Garmin to Grafana sync) *Jul 2026*

– Built a Python pipeline that syncs Garmin data into TimescaleDB for Grafana dashboards
– Enabled in-depth monitoring and visualization of fitness data

Museum Exhibit *Apr 2026*

– Built an interactive asteroid-impact exhibit for elementary-school visitors
– Detected impact location and magnitude using a gyroscope
– Sustained 99.5% availability with a portable, ruggedized design